1.26 Exponential Growth and Decay

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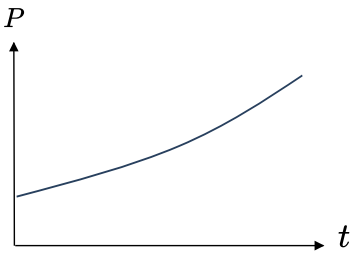
Exponential Modelling

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There are two key features of exponential functions which make them suitable for **population growth**:

1. **$\mathbf{a^x}$ gets $\mathbf{a}$ times bigger each time $\mathbf{x}$ increases by 1. (Because $\mathbf{a^{x+1}=a\times a^x}$)**
   With population growth, we typically have a fixed percentage increase each year. So suppose the growth was 10% a year, and we used the equivalent decimal multiplier, 1.1, as a. Then 1.1^t, where t is the number of years, would get 1.1 times bigger each year.
2. **The rate of increase is proportional to the size of the population at a given moment.**
   This makes sense: The 10% increase of a population will be twice as large if the population itself is twice as large.
3.

Suppose the population P in *The Republic of Nqmir* is modelled by $P = 100e^{3t}$ where t is the numbers years since *The Republic* was established.



What is the initial population?

What is the initial rate of population growth?

"Use of Technology" I use the $e^{\square}$ key. I can also use [ALPHA] [e] to get e without a power.

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Another Example

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[Textbook] The density of a pesticide in a given section of field, P mg/m^2, can be modelled by the equation $P = 160e^{-0.006t}$
where t is the time in days since the pesticide was first applied.
a. Use this model to estimate the density of pesticide after 15 days.
b. Interpret the meaning of the value 160 in this model.
c. Show that $\frac{dP}{dt} = kP$, where k is a constant, and state the value of k.
d. Interpret the significance of the sign of your answer in part (c).
e. Sketch the graph of P against t.

a

b

c

$\frac{dP}{dt} = (160 \times -0.006)e^{-0.006t}$

d

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Exercise 14C (Pages 318-319)

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5 On Earth, the atmospheric pressure, p, in bars can be modelled approximately by the formula $p = e^{-0.13h}$ where h is the height above sea level in kilometres.

a Use this model to estimate the pressure at the top of Mount Rainier, which has an altitude of $4.394\,\text{km}$. \hfill (1 mark

b Demonstrate that $\frac{\mathrm{d}p}{\mathrm{d}h} = kp$ where k is a constant to be found. \hfill (2 marks

c Interpret the significance of the sign of k in part **b**. \hfill (1 mark

d This model predicts that the atmospheric pressure will change by $s\%$ for every

Km gained in height. Calculate the value of s.

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Exercise 14C (Pages 318-319)

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6 Nigel has bought a tractor for £20 000. He wants to model the depreciation of the value of his tractor, £T, in t years. His friend suggests two models:

Model 1: $T = 20\,000e^{-0.24t}$

Model 2: $T = 19\,000e^{-0.255t} + 1000$

a Use both models to predict the value of the tractor after one year. Compare your results. \hfill (2 mark

b Use both models to predict the value of the tractor after ten years. Compare your results. \hfill (2 mark

c Sketch a graph of T against t for both models. \hfill (2 mark

d Interpret the meaning of the 1000 in model 2, and suggest why this might make model 2 more realistic. \hfill (1 mar